

Lecture 6

EE 421 / CS 425

Digital System Design

Fall 2026
Shahid Masud

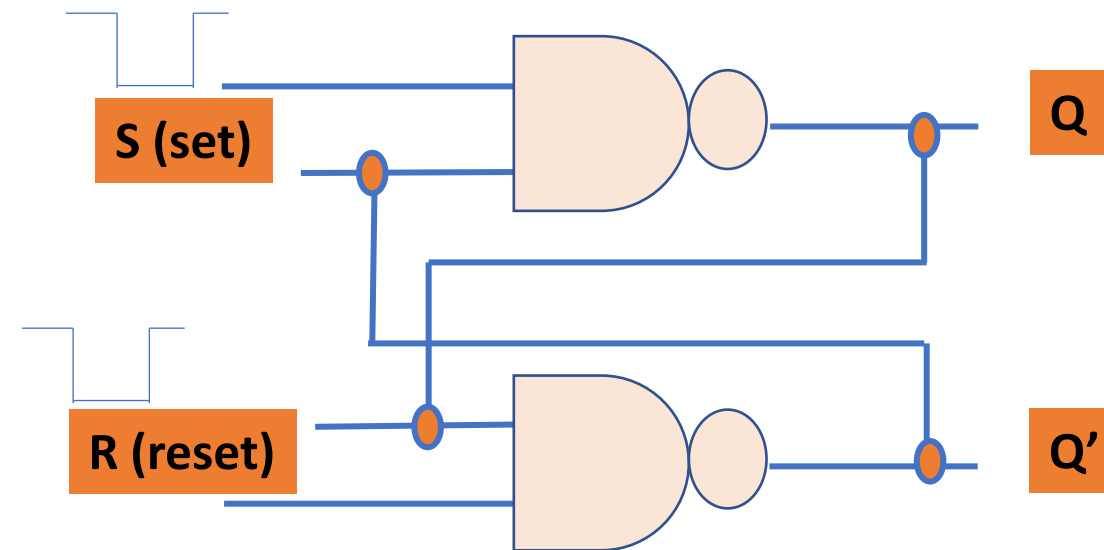
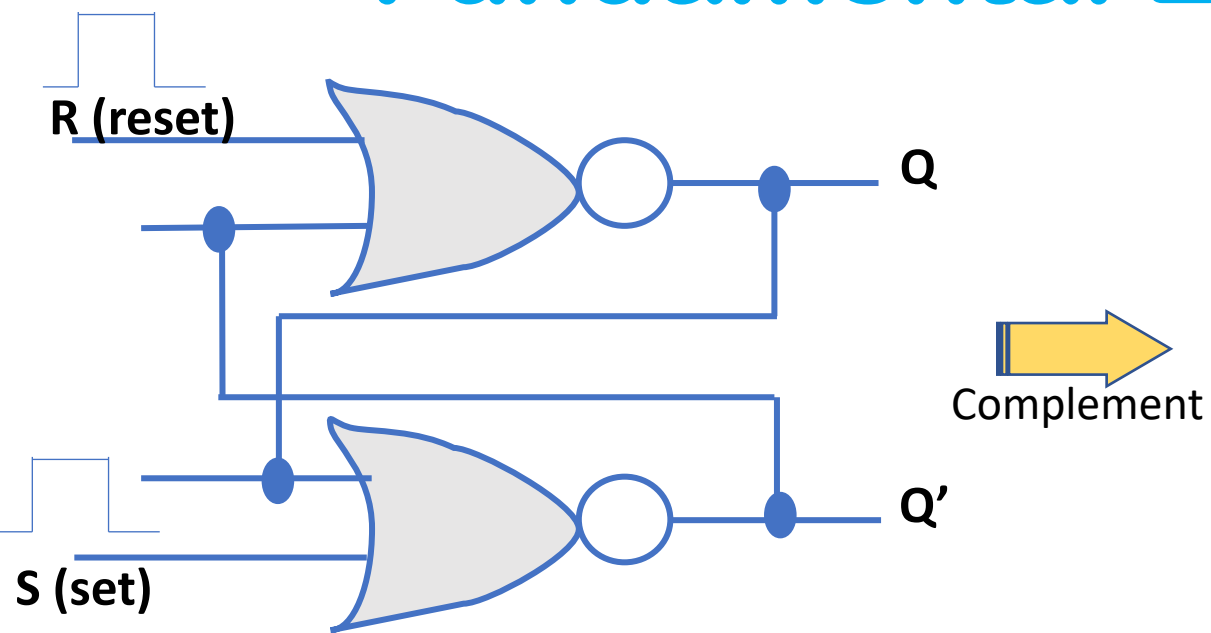
Topics

- Review of Sequential Digital Logic Circuits
- Basics of Latches
- Basics of Flipflops
- Setup and Hold Times
- D Flipflop
- Clock Issues

Latches and Flipflops - Definitions

- **Latches:** Level Sensitive storage elements; that can transition at any point in time
- **Flipflops:** Edge Sensitive storage elements, that transition at clock edges, either rising or falling
- **Asynchronous Sequential Logic:** Based on latches and un-clocked transitions
- **Synchronous Sequential Logic:** Based on flipflops and only clocked transitions

Fundamental Element – SR Latch



SR Latch Using 2 Input NOR Gate – Function Table

S (Set)	R (Reset)	Q	Condition
0	0	Q (=1 after S=1, R=0)	Hold
0	0	Q (=0 after S=0, R=1)	Hold
0	1	0	Reset
1	0	1	Set
1	1	0	Not Allowed

RACE
Condition

SR Latch Using 2 Input NAND Gate – Function Table

S	R	Q	Condition
0	0	1	Not Allowed
0	1	1	Set
1	0	0	Reset
1	1	Q (=0 after S=1, R=0)	Hold
1	1	Q (=1 after S=0, R=1)	Hold

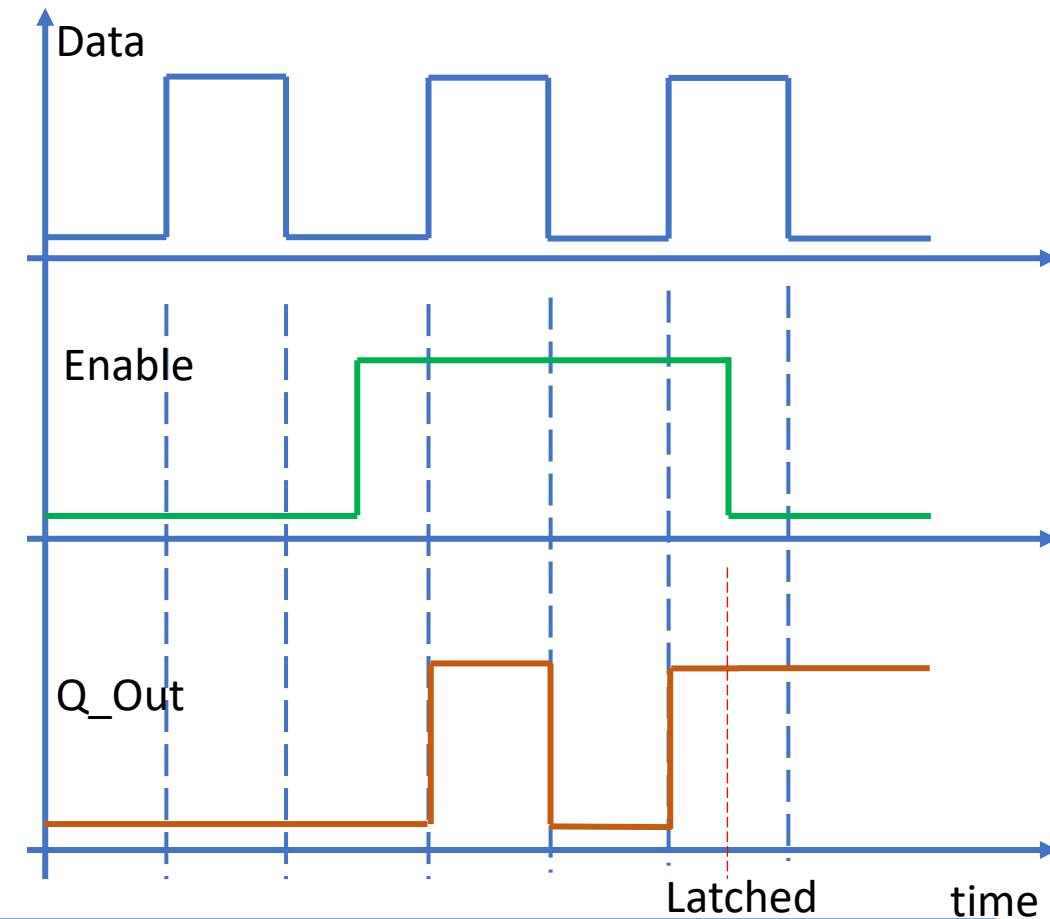
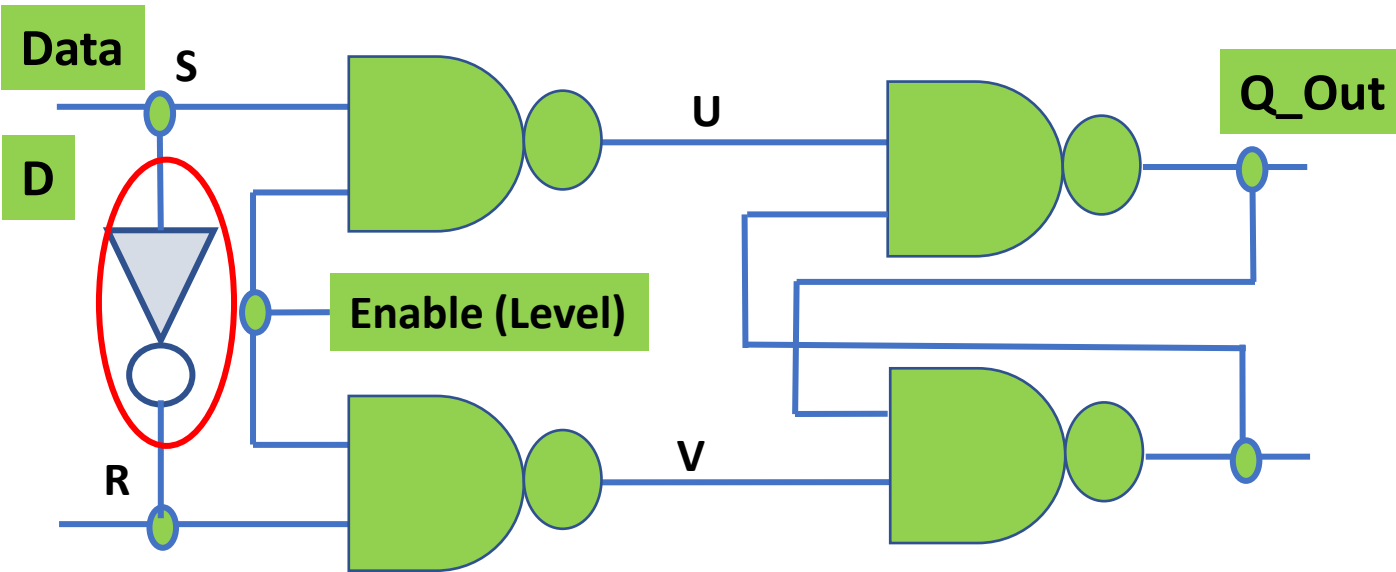
RACE
Condition

Transparent Latch

- An additional 'Enable' input is provided
- Definition: Output of Transparent Latch changes in response to the data input only when the Latch is 'Enabled'.
- Changes in input are straight away visible at the output

Transparent Latch Circuit

Timing Diagram – With Input Inverter



SR Latch With Enable (Without Inverter)

En	S	R	Next State of Q
0	X	X	No Change
1	0	0	No Change
1	0	1	Q=0; Reset State
1	1	0	Q=1; Set State
1	1	1	Not Allowed

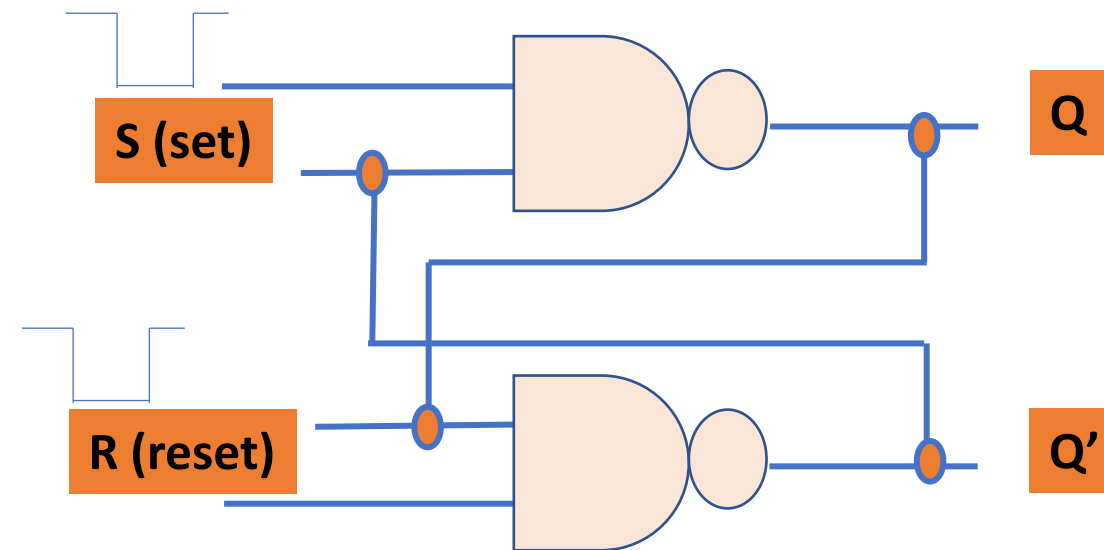
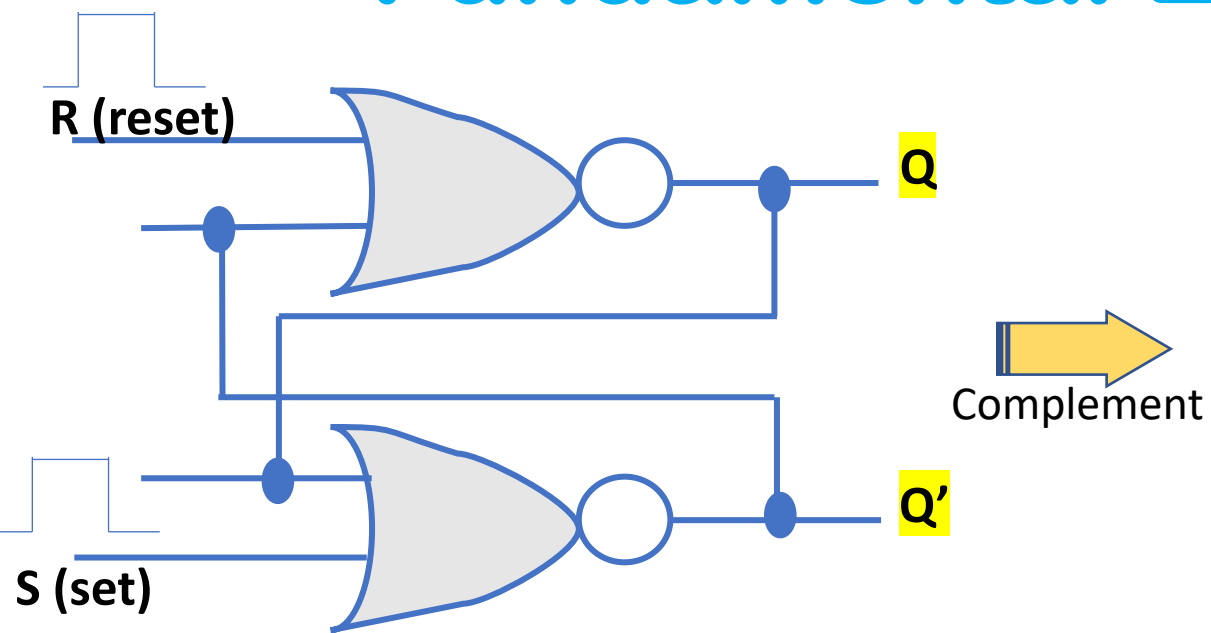
When Enable is '0'; U='1' and V='1'; Hold Condition in SR Latch

Only these two states are valid when inverter is inserted

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SR Latch Using 2 Input NAND Gate – Function Table

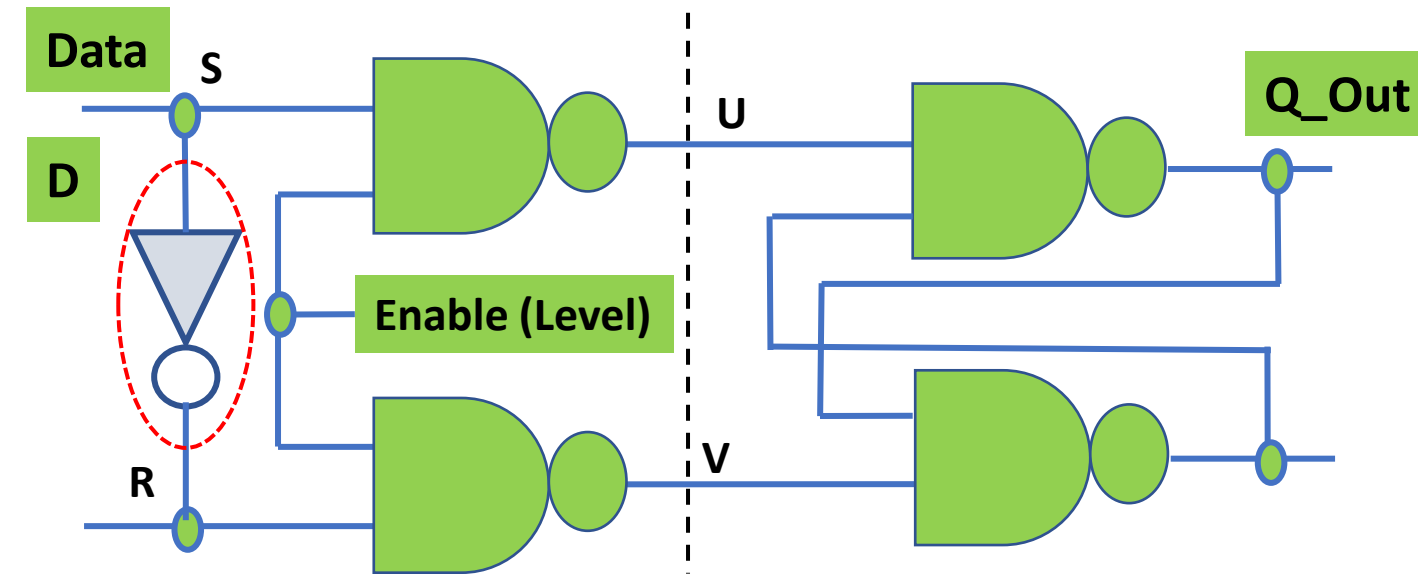
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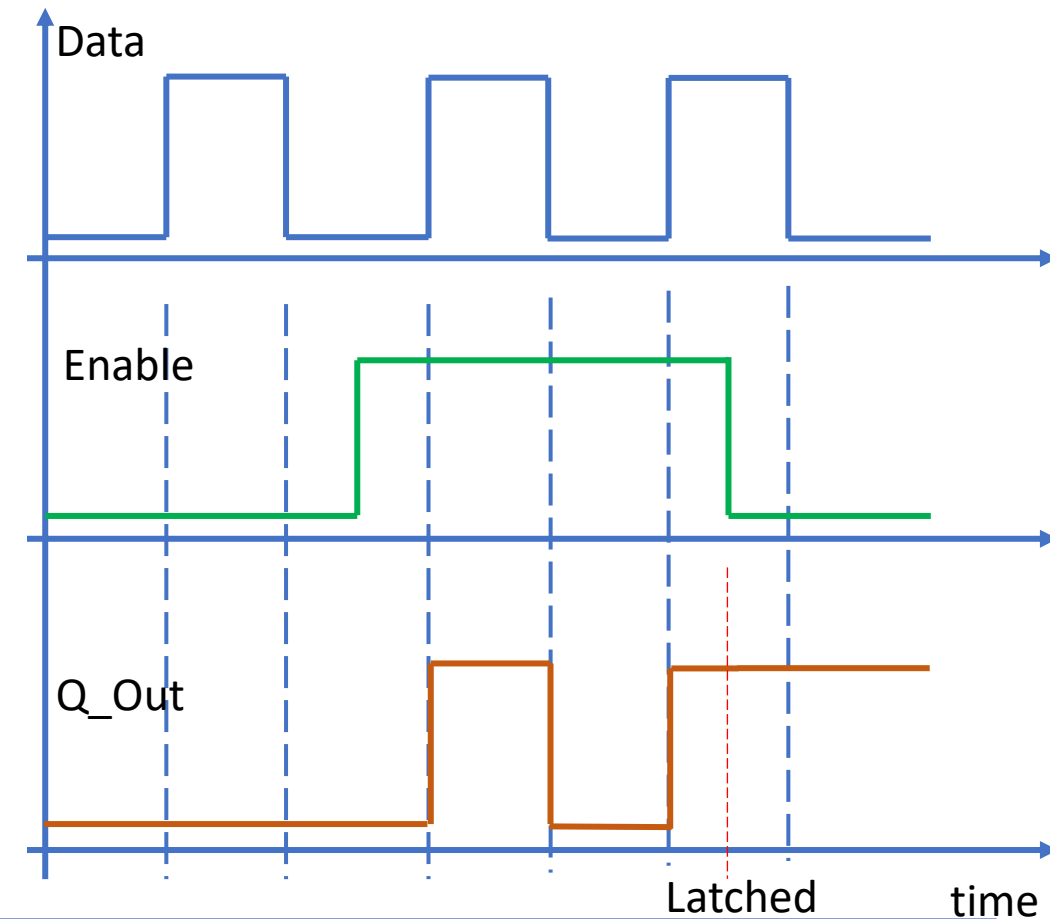
Transparent Latch Circuit

Timing Diagram – With Input Inverter



SR Latch With Enable (Without Inverter)

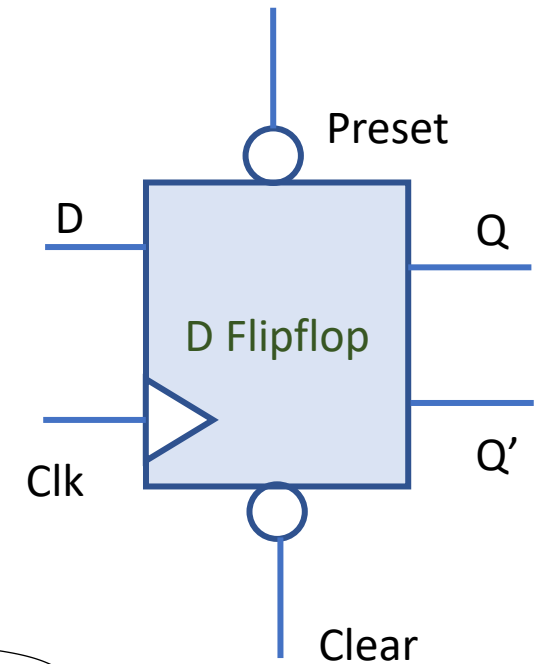
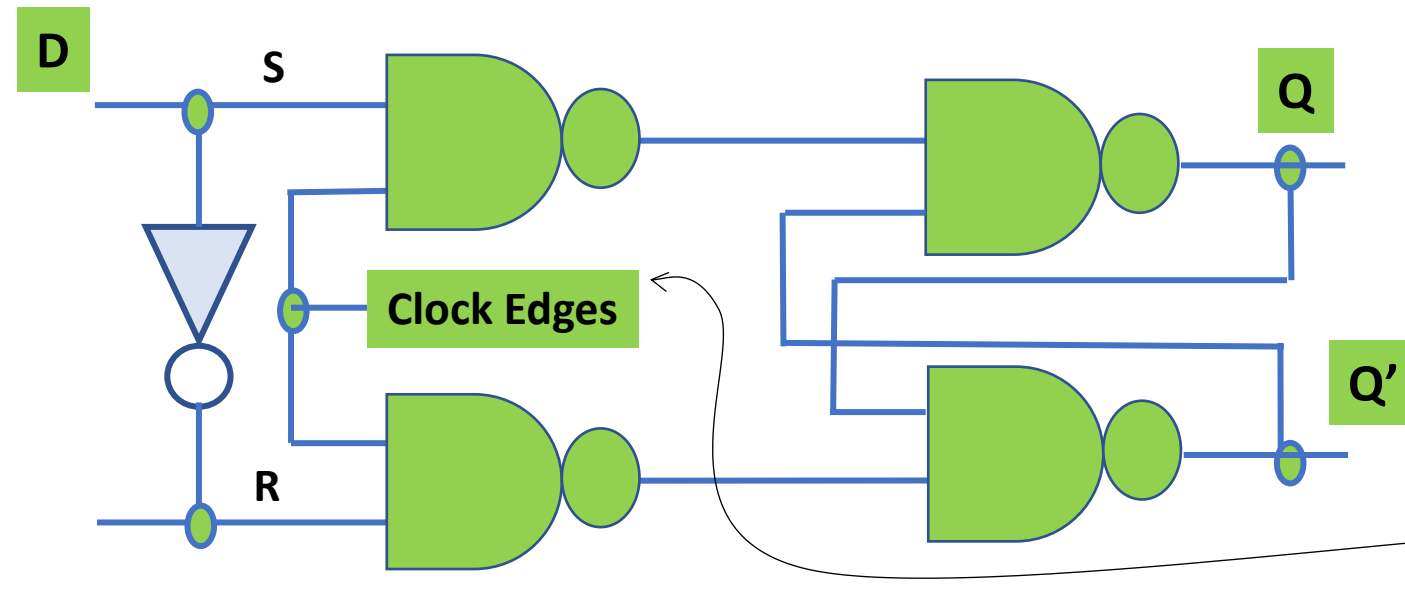
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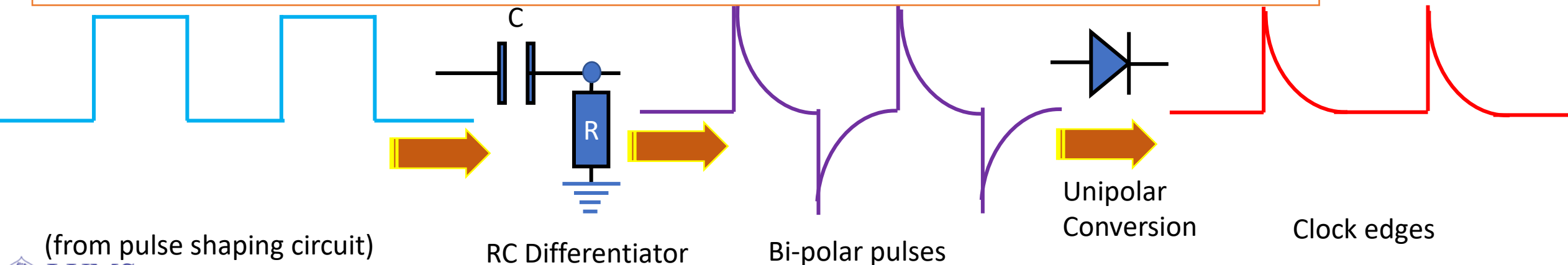
When Enable is '0'; U='1' and V='1'; Hold Condition in SR Latch

Only these two states are valid when inverter is inserted

D Flipflop from Transparent Latch



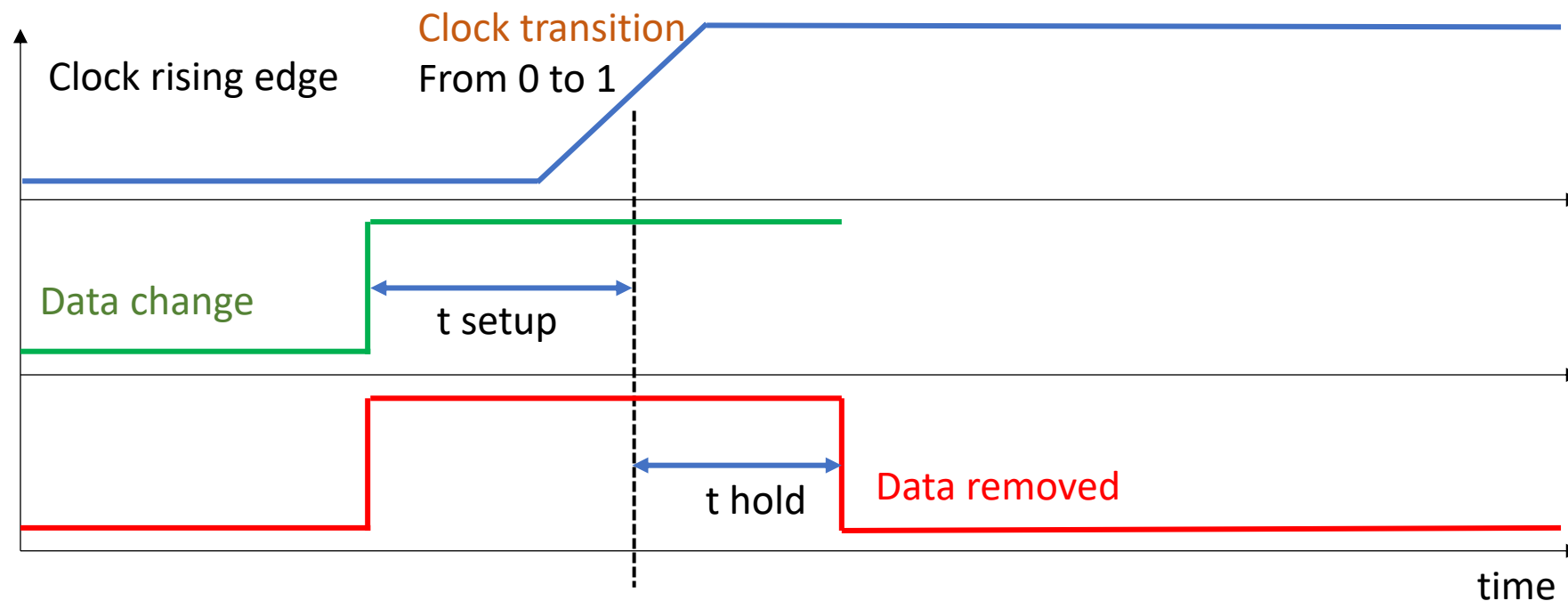
NOTE: Usually a Master-Slave Arrangement is used where output is available after clock goes 0 to 1 to 0



Timing Definitions

- **Setup Time:** The duration of time the Data needs to be present before the arrival of clock.
- Any Data activity, in tandem with clock edge, violates setup time and will not be recorded in the D flipflop
- **Hold Time:** The time for which data must remain stable after the removal of clock edge.
- If the Data changes too soon, the internal circuit may not have settled with the new value.

Setup and Hold Times

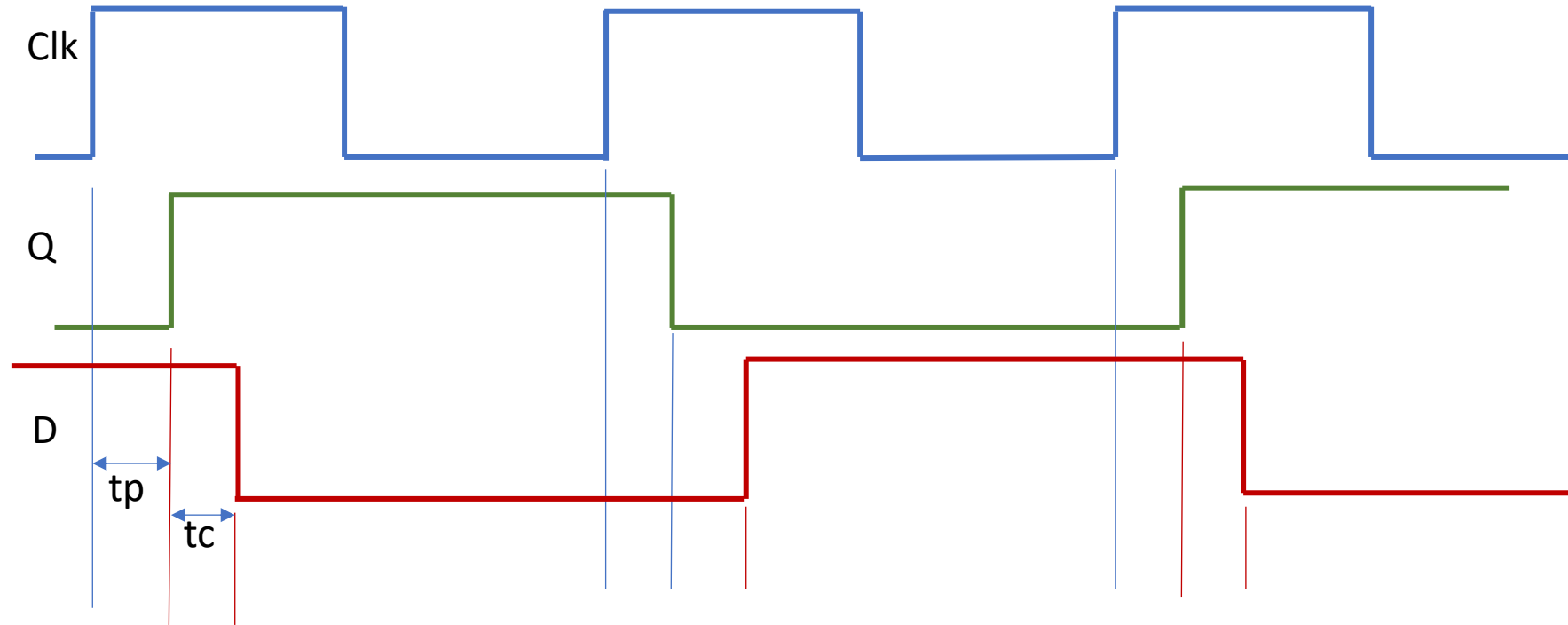
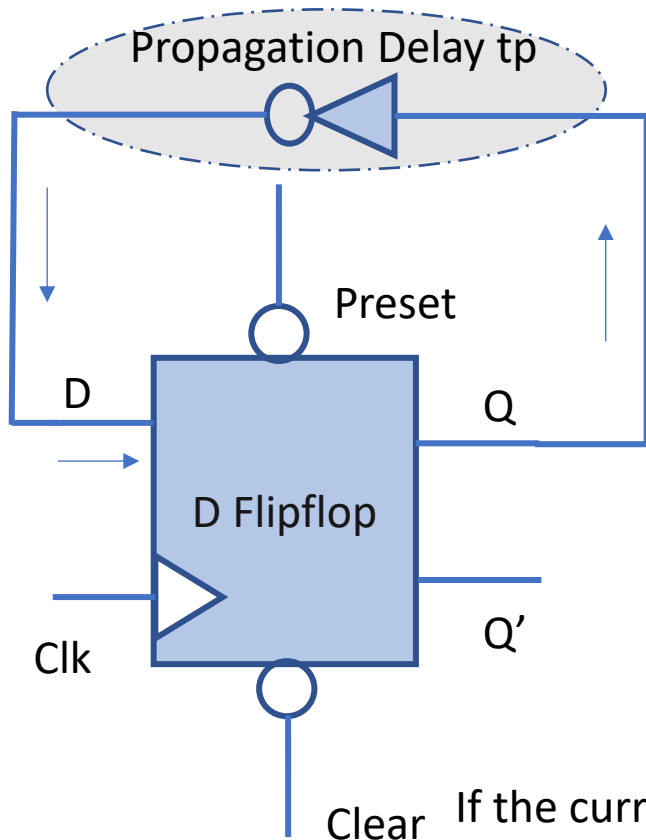


Maximum clock frequency has to be slower than $(t_{\text{setup}} + t_{\text{hold}})$ to maintain correct data

These timings needed both ways, slow data and fast clock; as well as slow clock and fast data

Timing in a simple frequency divider

For maximum clock speed consider → setup time, hold time, relevant propagation delays



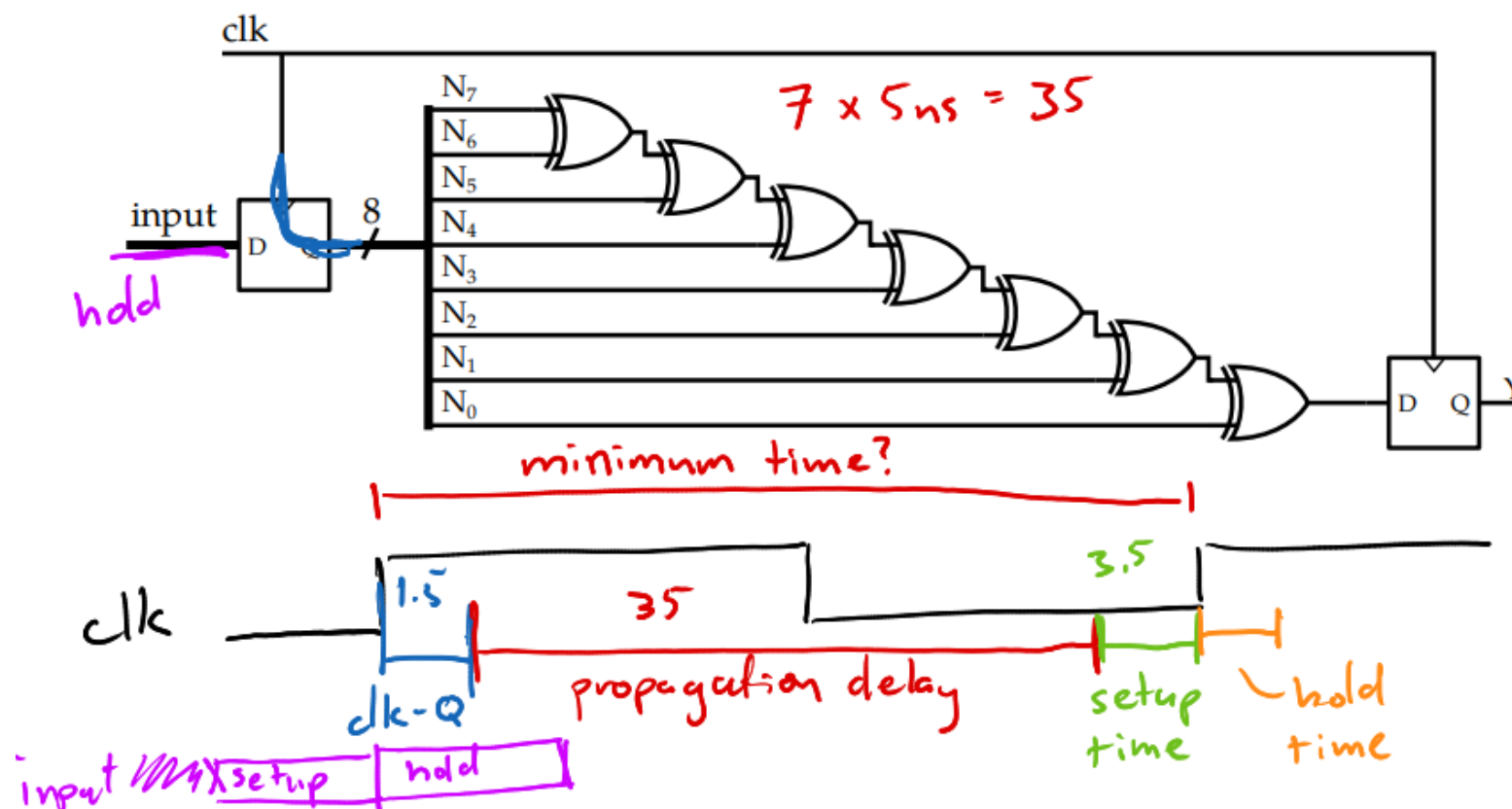
If the current output of flipflop is 1, a value of 0 will appear at D after propagation delay of inverter
 Assuming that next active edge of clock arrives after setup time has elapsed, the output of flipflop will go to 0
 A continuous waveform with period twice the period of Clk is observed at Q

Example of time calculation

How fast can this run?

Don't we care about time to result?

(i.e., what is the maximum clock frequency before timing failure?)



Gate	t_{pd} (ns)	t_{cd} (ns)
2-input XOR	5.0	1.7
clk-Q	1.5	0.75

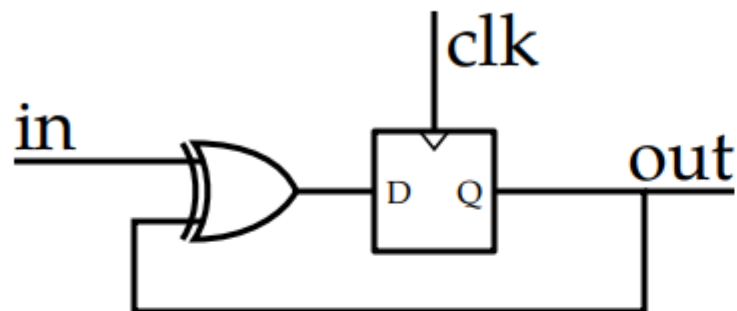
DFF setup time 3.5 ns
DFF hold time 2.0 ns

book value?

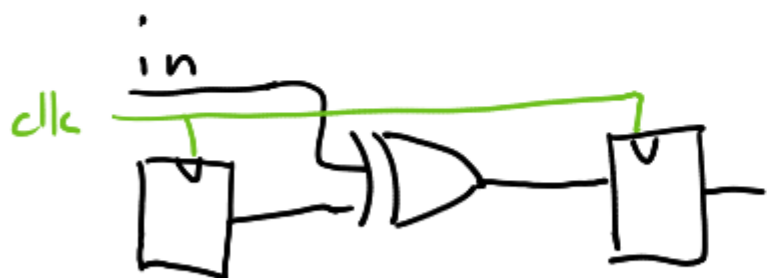
Challenge: what would this look like if you implemented it with 4-input LUTs?
How fast would it run?

Another Example of time calculation

How fast can this run?

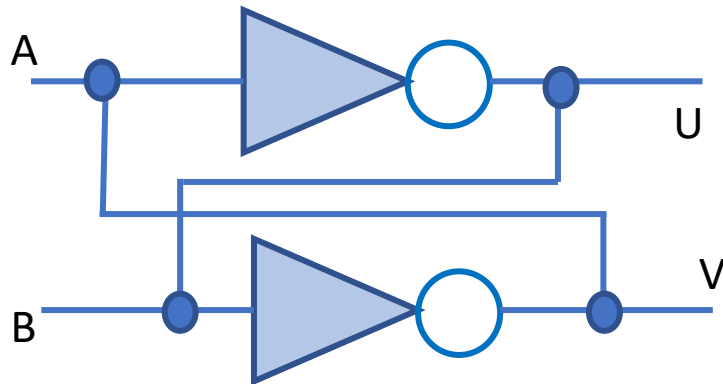


Gate	t_{pd} (ns)	t_{cd} (ns)
2-input XOR	5.0	1.7
clk-Q	1.5	0.75
DFF setup time	3.5 ns	
DFF hold time	2.0 ns	



$$\begin{array}{rclcl}
 \text{clk-Q} & + & \text{XOR} & + & \text{setup} \\
 1.5 & & 5 & & 3.5 \\
 & & & = & 10\text{ns} = 100\text{MHz}
 \end{array}$$

Metastable State



Imagine a circuit like this

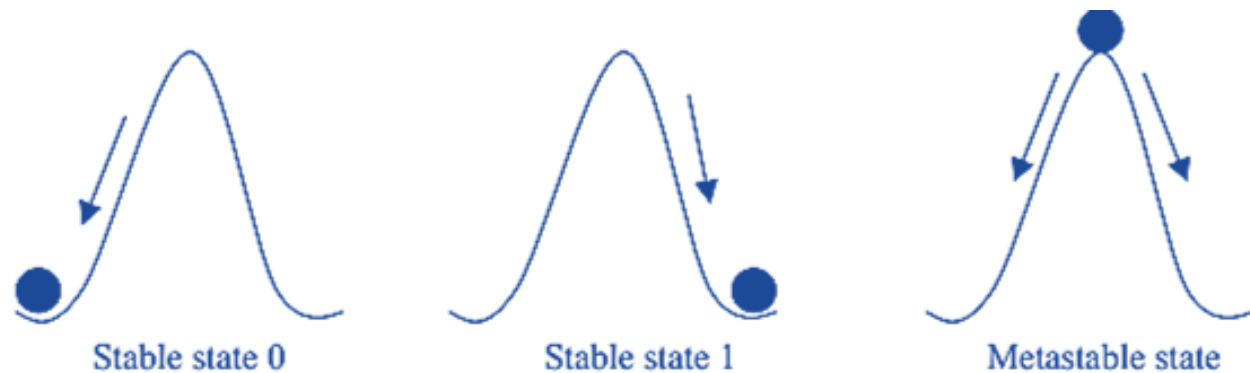


Given a pulse input:

Output reaches a value (or oscillates) somewhere in-between '0' and '1' and stays there, neither '0' nor '1', for some noticeable time

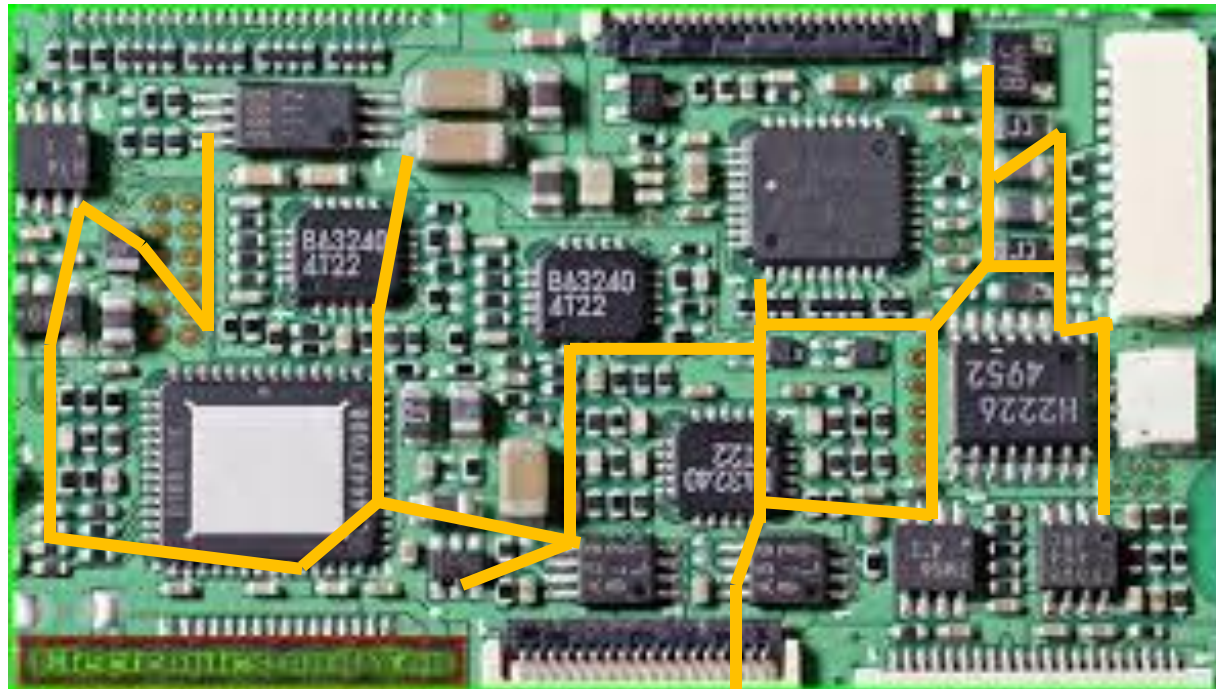
Why?

- Too small pulse violates setup and hold times
- Too less drive strength available to overcome the intermediate state



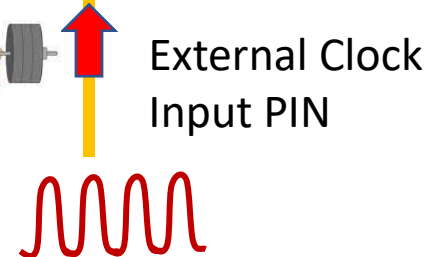
Clock - Problems

Clock is an essential
Signal in Synchronous
Digital Systems



Issues:

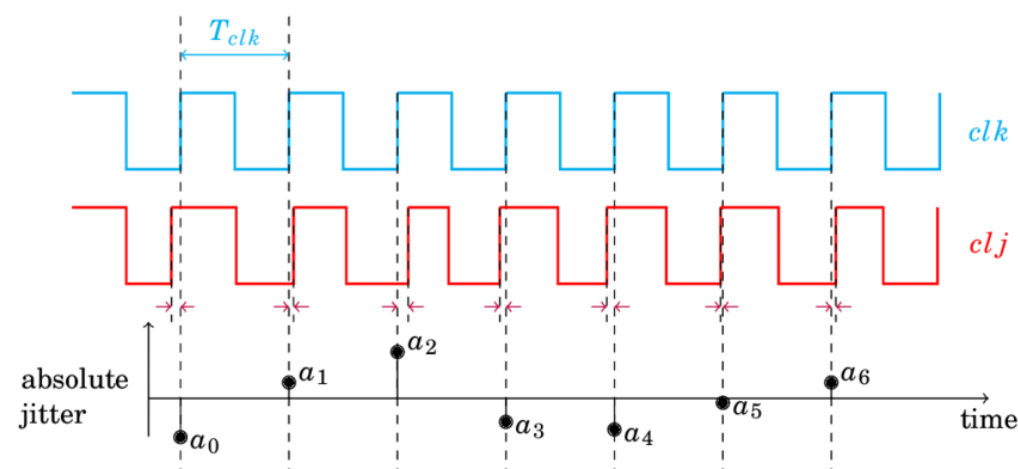
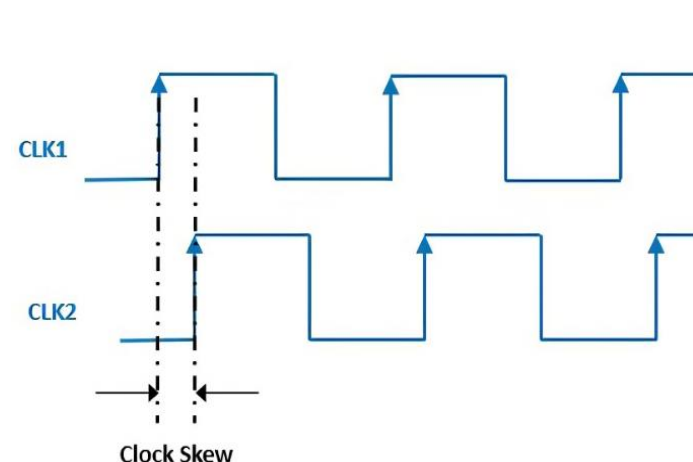
1. Too much load on Clock Input Pin
2. Variable path delay to Different Chips in circuit
3. Path is prone to picking up noise from surrounding



Clock Distribution Circuit

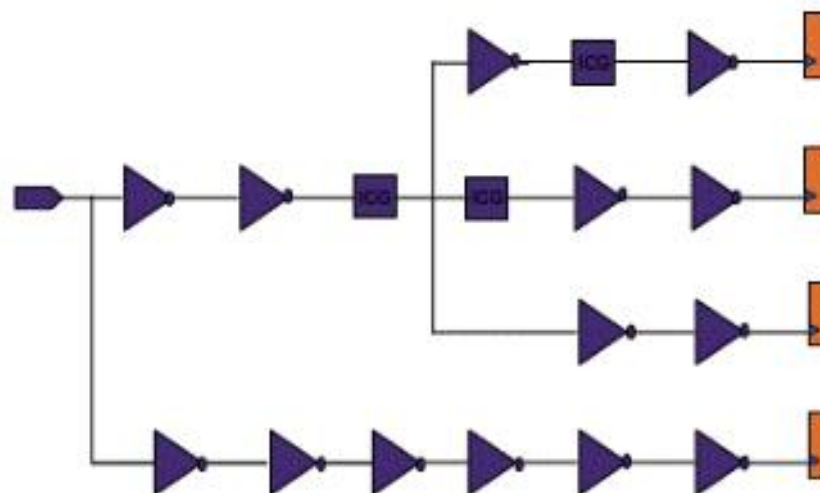
Qualities of a Good Clock Signal:

- Clock has to reach each and every module of a complex circuit
- Clock needs very high signal drive strength
- Clock Edge has to appear at the same instant across all modules (**skew**)
- Clock should be **stable** and free from **jitter** (time variations along stable reference point)



Clock Tree Synthesis

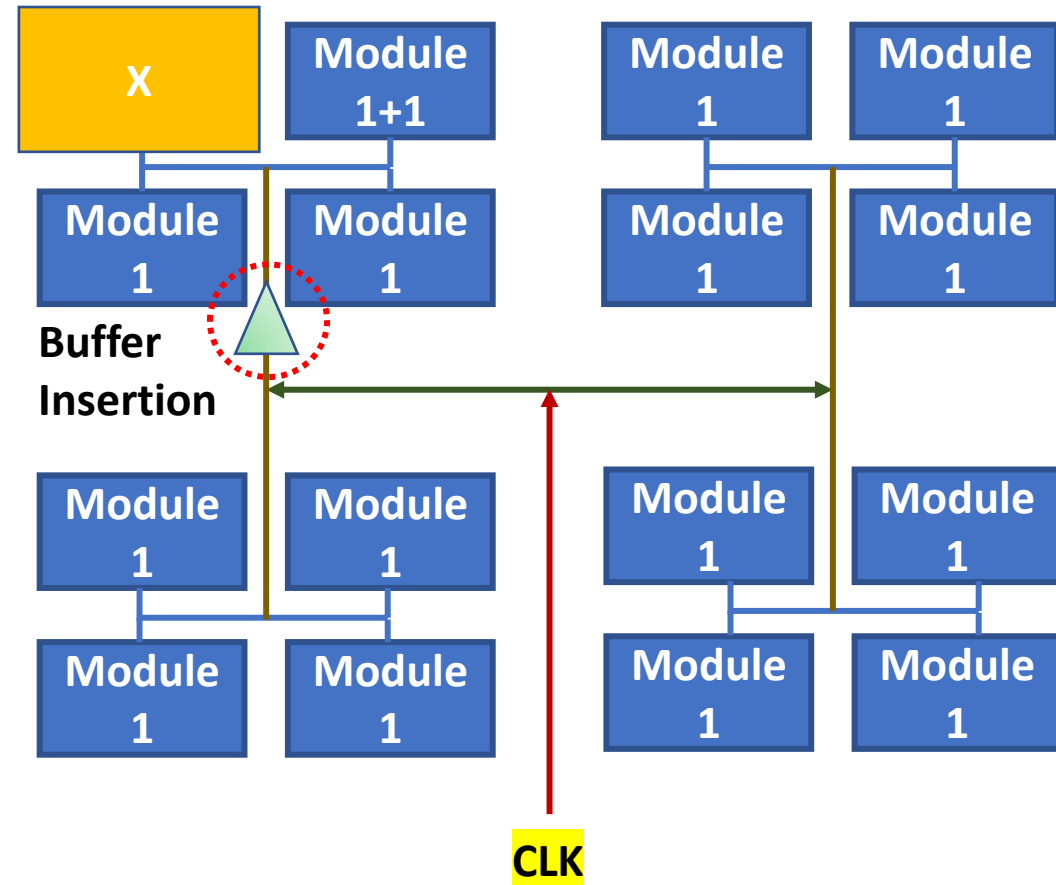
- Clock tree synthesis is performed during the physical design process considering the effects of place and route, channel impedance, parasitic loads, etc.
- Then through the insertion of buffers or inverters along the clock paths, it minimize or balances skew of important clock signal chains, build a clock tree that achieves proper timing across the entire design.



Ref: <https://www.system-to-asic.com/blog/what-is-asic-design/>

Some Techniques for Clock Distribution

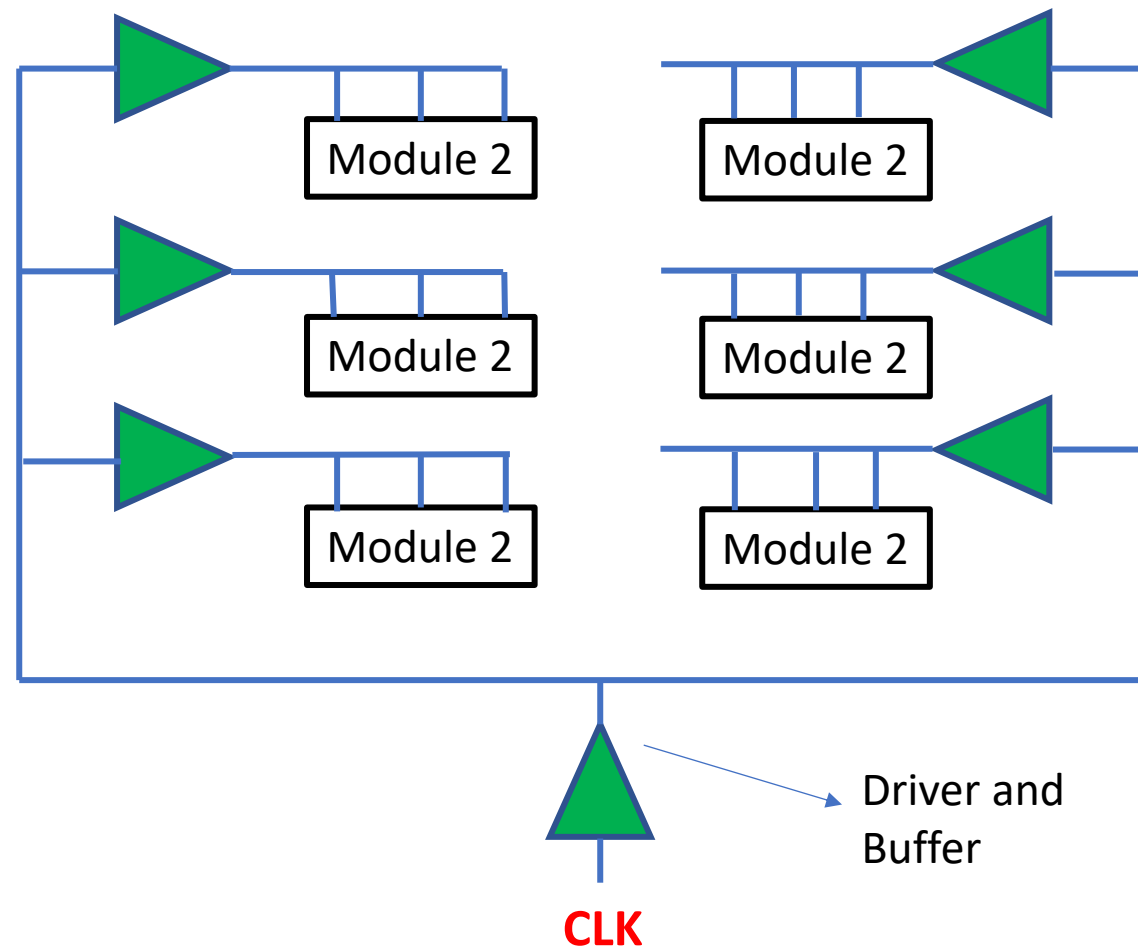
H-Tree Clock Synthesis
Path is balanced through
Tree structure



H Tree Clock Distribution Network
Buffer Insertion for Delay Balancing in un-balanced trees

Hierarchical Clock

**Buffer based Path
Delay Balancing**



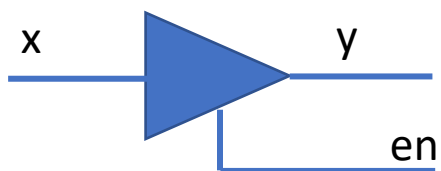
Two Level Distributed Clock Buffering

Special Purpose Combinational Circuits

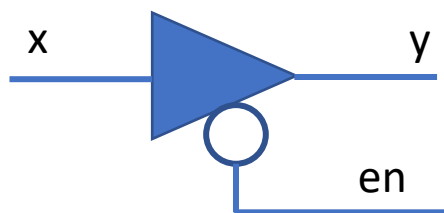
Three State Buffers for Processor Buses

- Buses are multi-wire signal paths that connect multiple functional units in a computer system, e.g., Address bus, peripheral bus, etc.
- Simplifies design and architecture of a computer system
- Tradeoff is that the access to the bus has to be managed to avoid conflicts
- Three state hardware devices provide a dynamic interface between a bus and a circuit, providing signals when enabled else an open circuit
- The input goes to the output of the tri-state buffer when enable signal is asserted

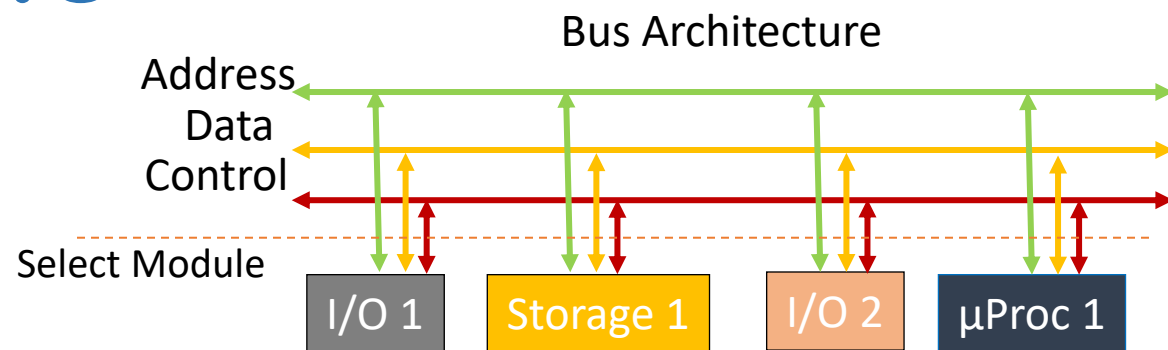
Three State Bus Drivers



x	en	y
0	0	Hi_Z
0	1	0
1	0	Hi_Z
1	1	1



x	en	y
0	0	0
0	1	Hi_Z
1	0	1
1	1	Hi_Z

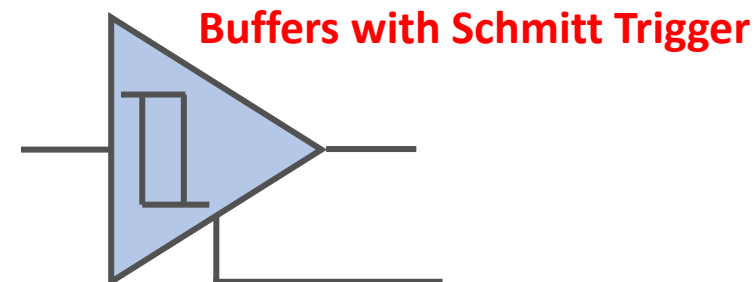


Typical Bus Structure utilizing 3-state buffers

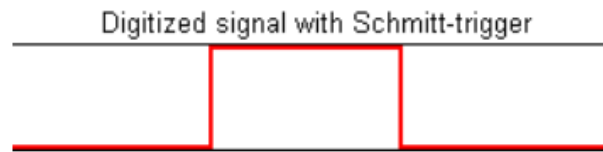
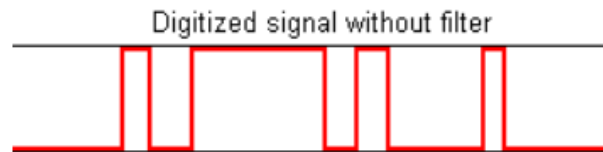
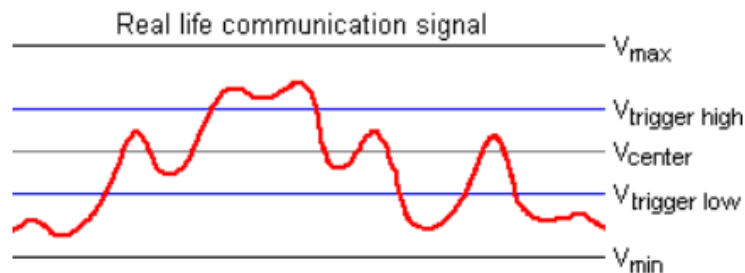
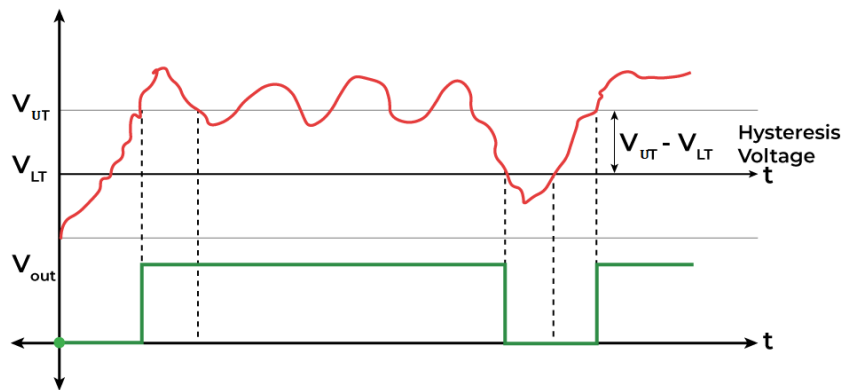
Inverting and Non-Inverting 3-State Buffers / Drivers Exist

The Enable can be active high or active low

Buffers with Hysteresis Loop are available for better noise immunity



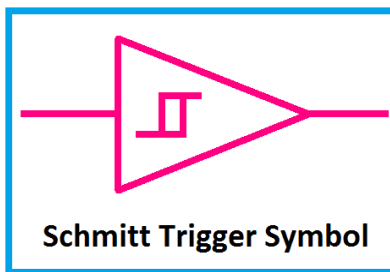
Schmitt Trigger



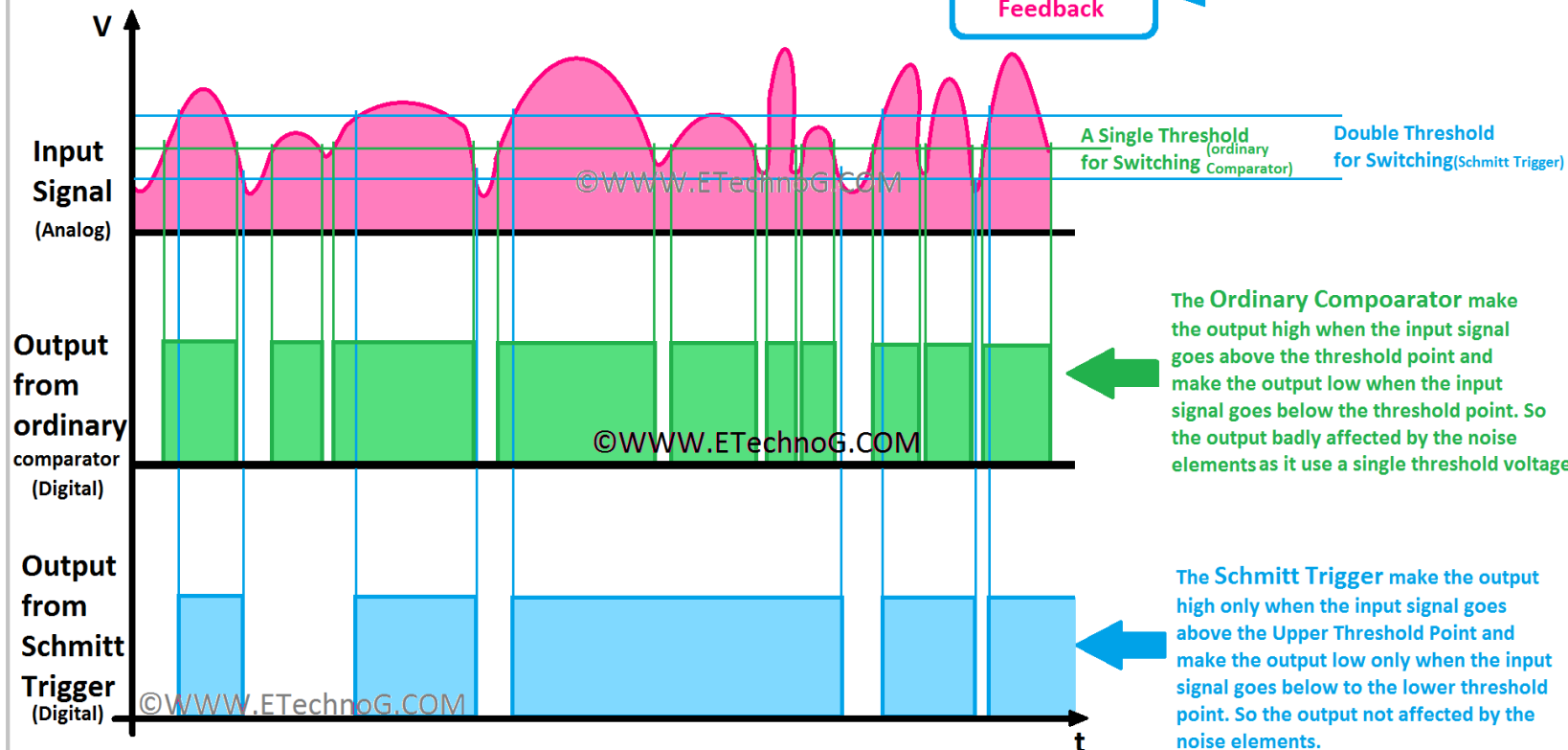
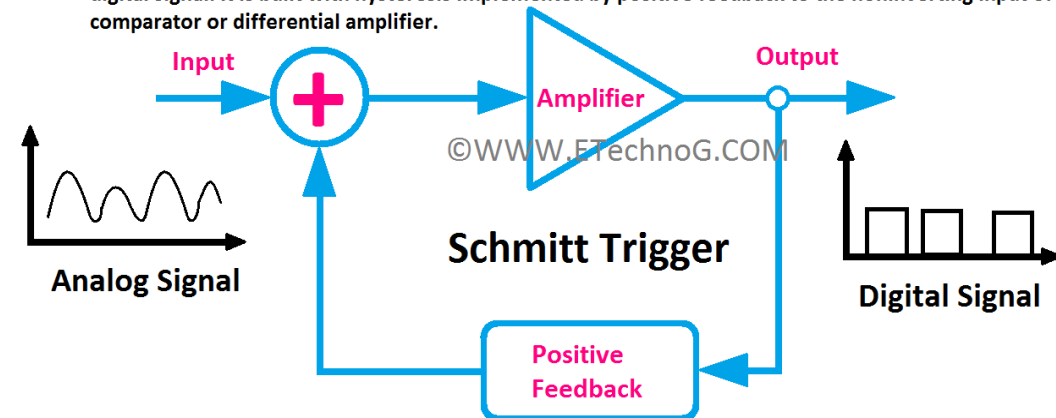
Schmitt trigger voltage levels

Figure. 01

How Schmitt Trigger Works?(Learn with Diagram)

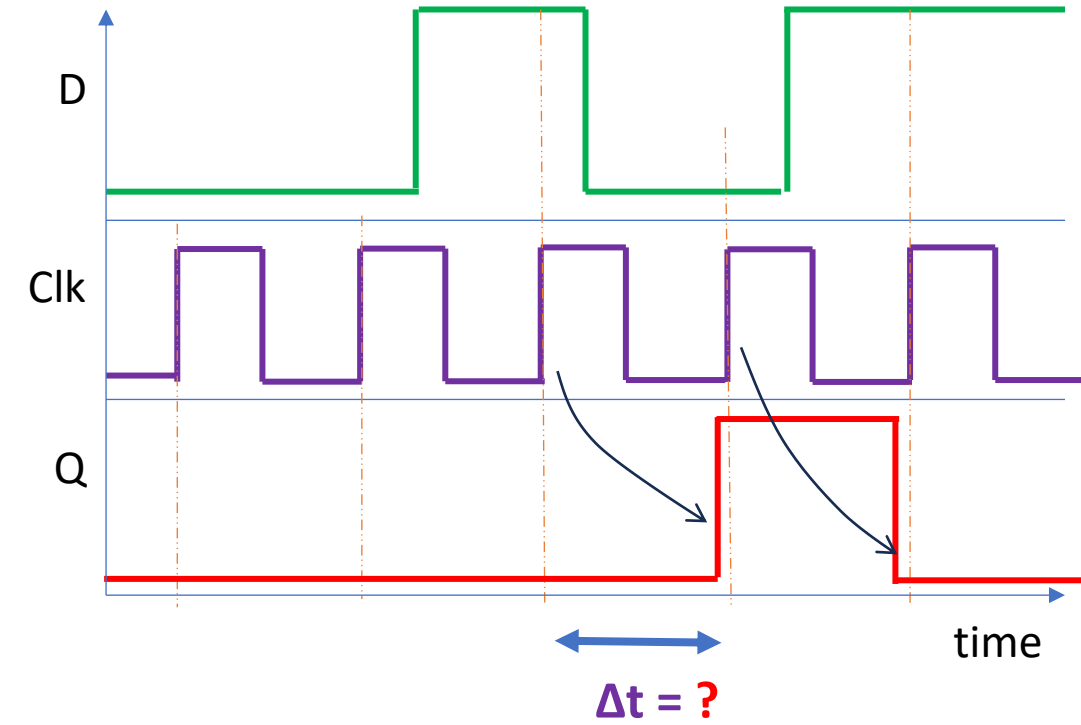
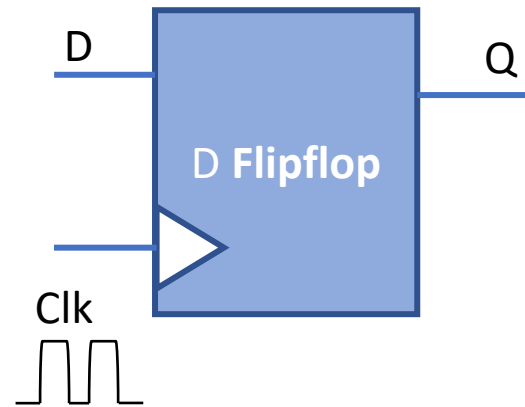
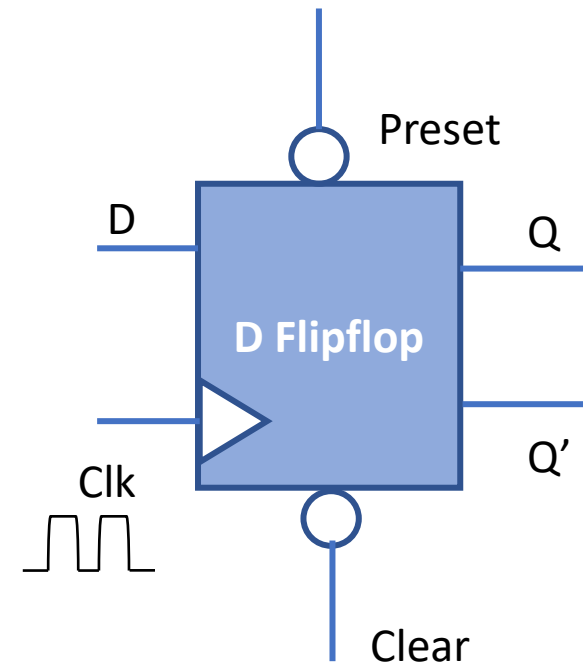


A Schmitt Trigger basically a Active Comparator circuit that is used to convert the analog signal into digital signal. It is built with hysteresis implemented by positive feedback to the noninverting input of a comparator or differential amplifier.



Sequential Elements and Circuits

D Flipflops – Behavior and Equation



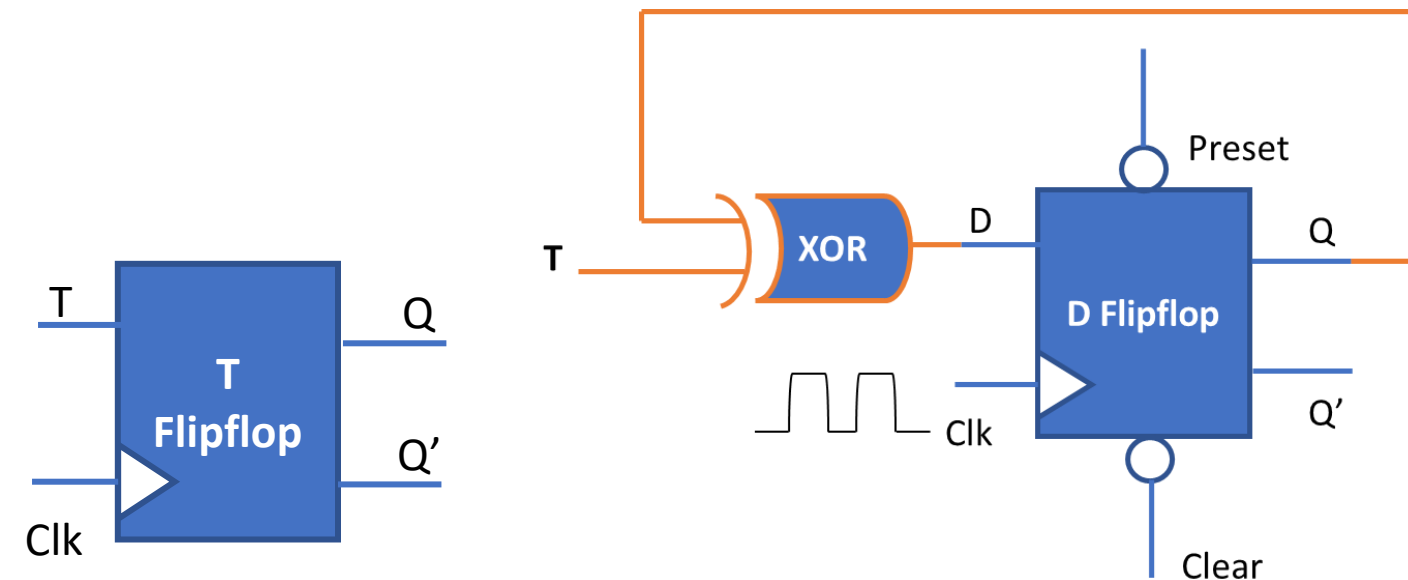
Characteristic Equation:

$$Q(t+1) = D$$

Q after clock pulse gets value of D

D	Q(t+1)
0	0; Reset
1	1; Set

T Flipflop – Behavior and Equation

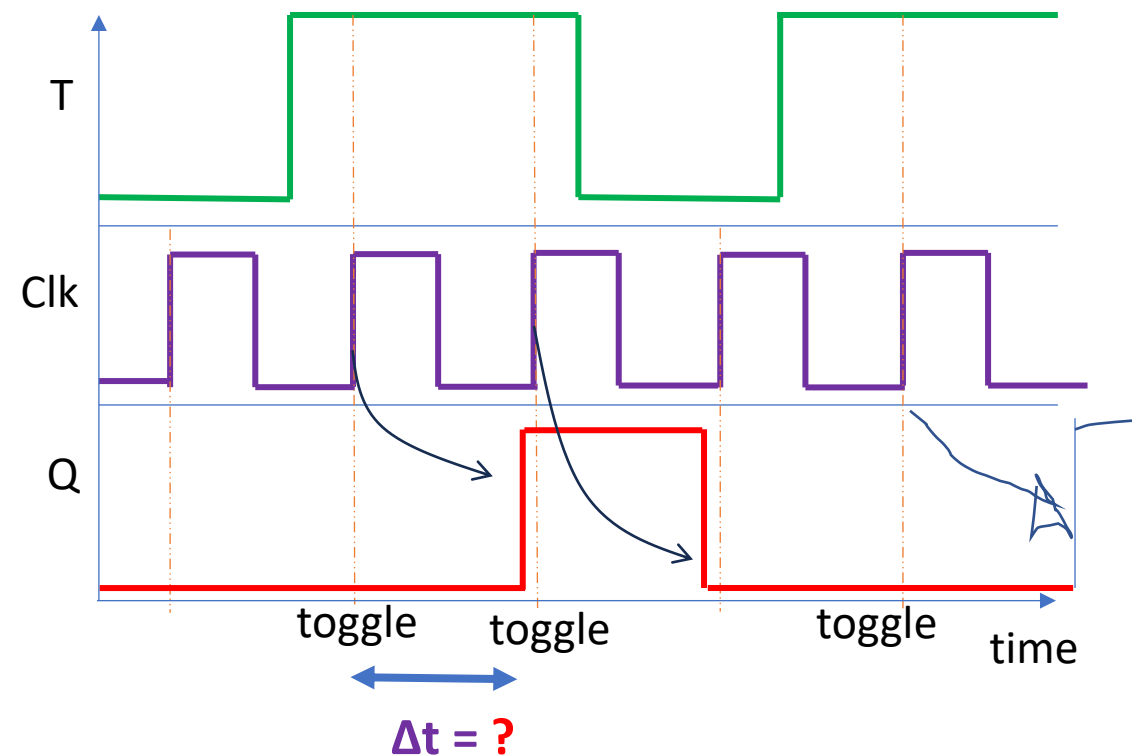


Characteristic Equation:

$$Q(t+1) = T \oplus Q$$

$$= T'Q + TQ'$$

T	Q(t+1)
0	Q(t); No change
1	Q'(t); Toggle or Complement



In this course,

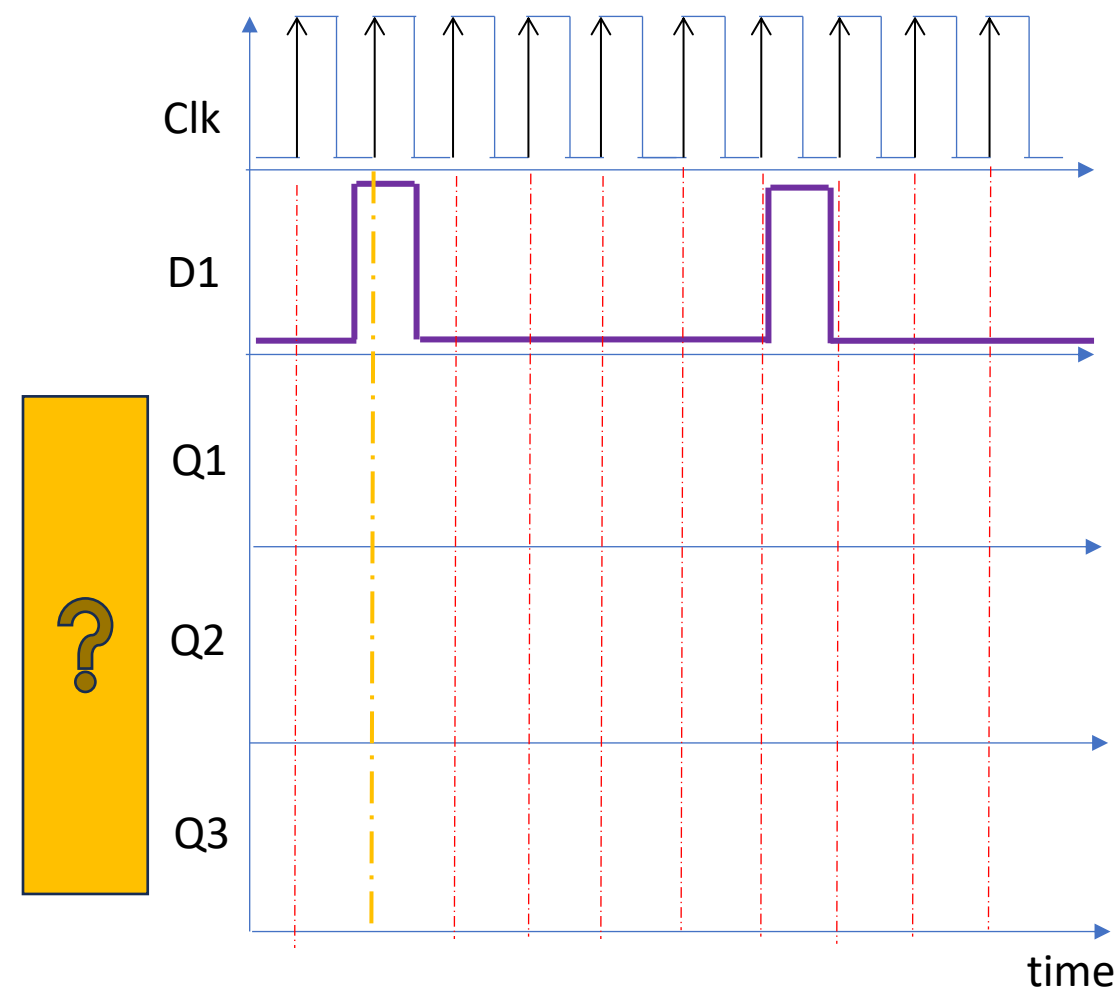
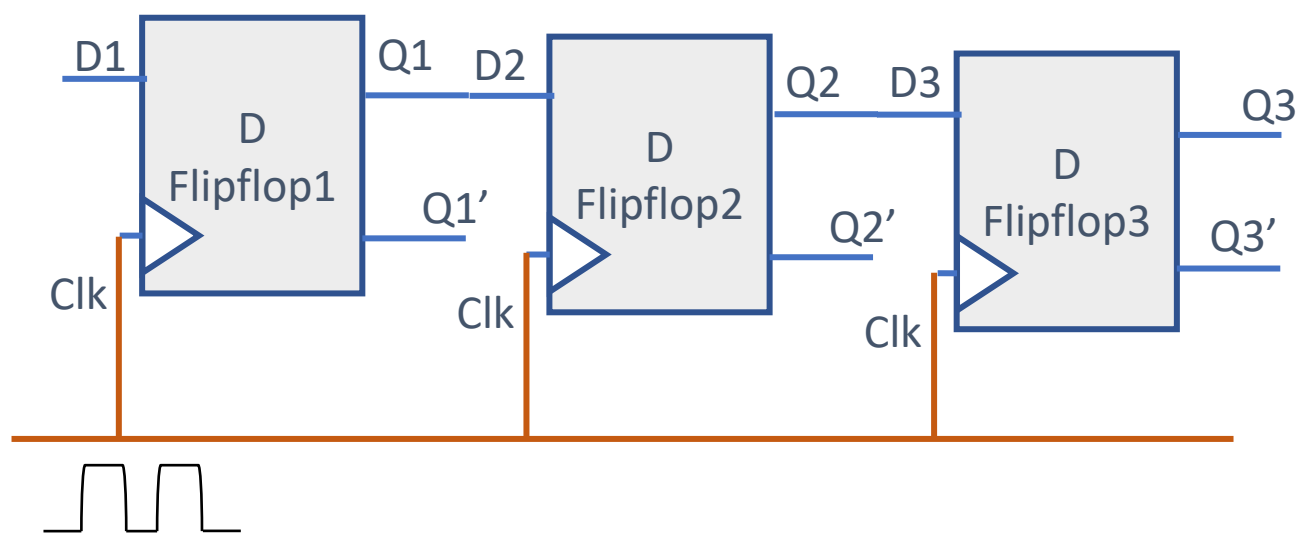
**We will mostly use D Flipflops
and sometimes T Flipflops**

We want to study complex and high-speed designs

Sequential Circuits: Counters, Sequencers, State Machines

- **Counters:** count in a fixed sequence, up or down
- **Sequencers:** sequence through a pre-defined arbitrary sequence
- **State Machines:** Sequential and Combinational Circuits work together to go one of many next states based on current inputs

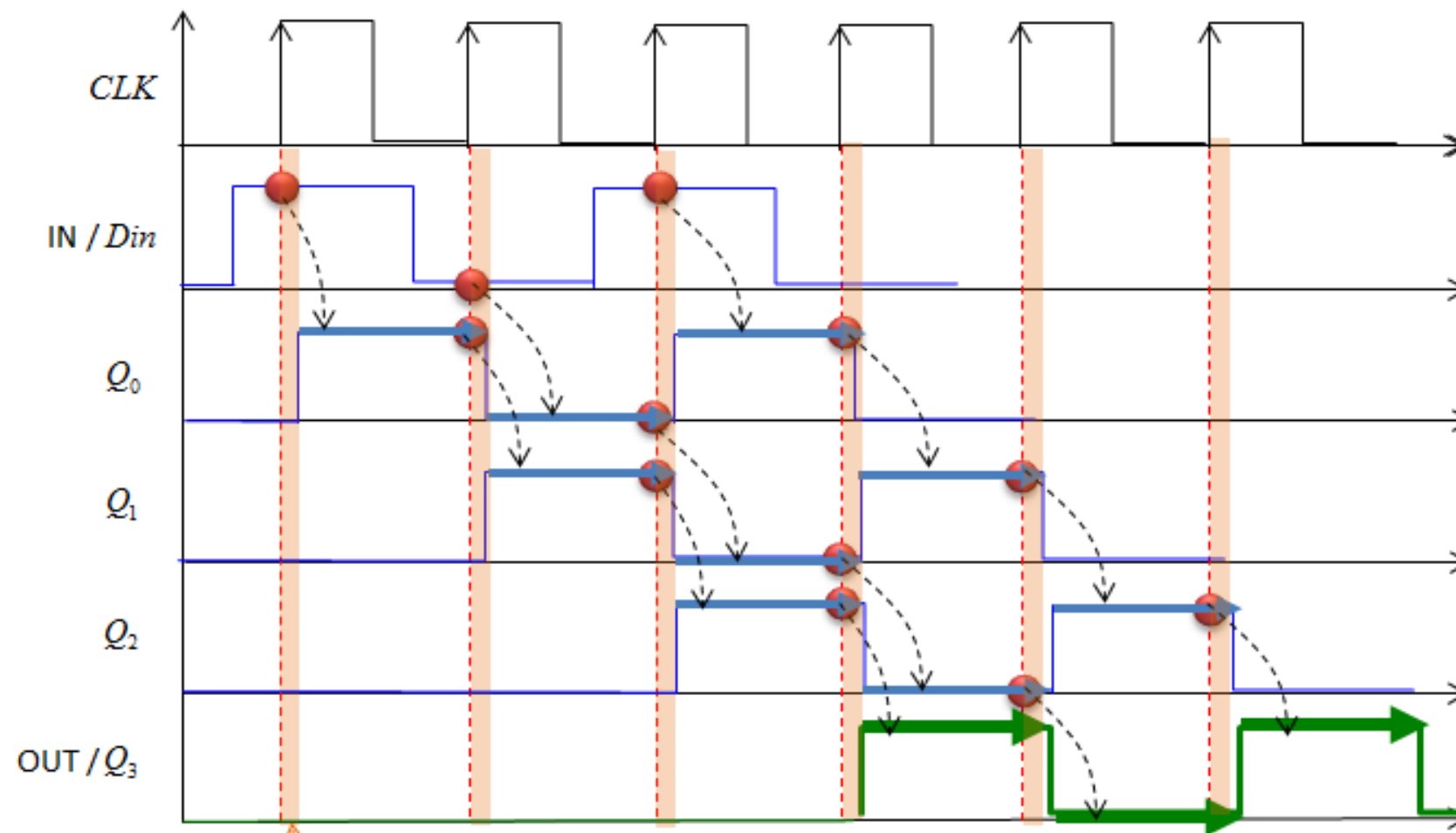
Moving Through Register Chain



Only 'EDGES' of Clocks are important for Flipflops

3 DFF Shift Register Timing Diagram from a website

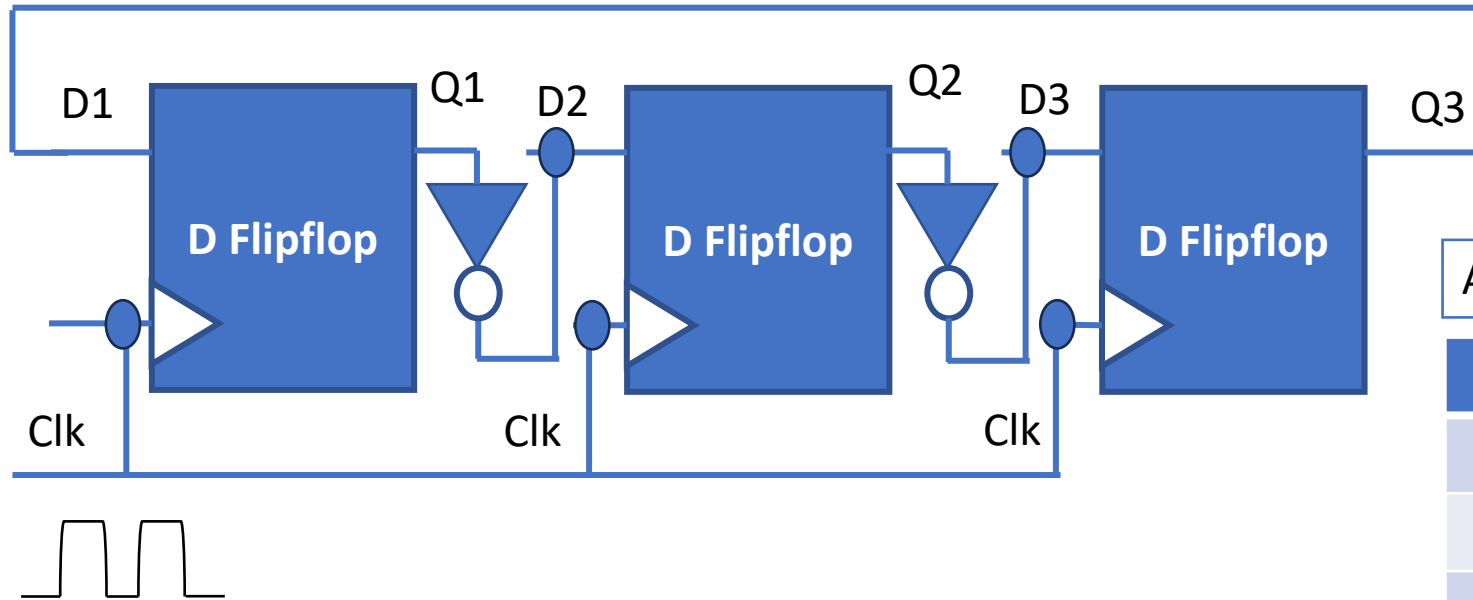
Look closely at the
Area close to
Clock transition,
Shaded pink lines



We assume that there is a small delay at Q state of each flipflop to make the timing diagram easily understandable.

But in reality, the situation would be a little more complicated than this simple time delay.

Cyclic Shift Register using D flipflops



Assume all DFFs are Cleared to **0** at Startup

Clock No.	Q1	Q2	Q3
1	0	0	0
2	0	1	1
3	1	1	0
4	0	0	0
5	0	1	1
6	1	1	0
7	0	0	0
CONTINUE			

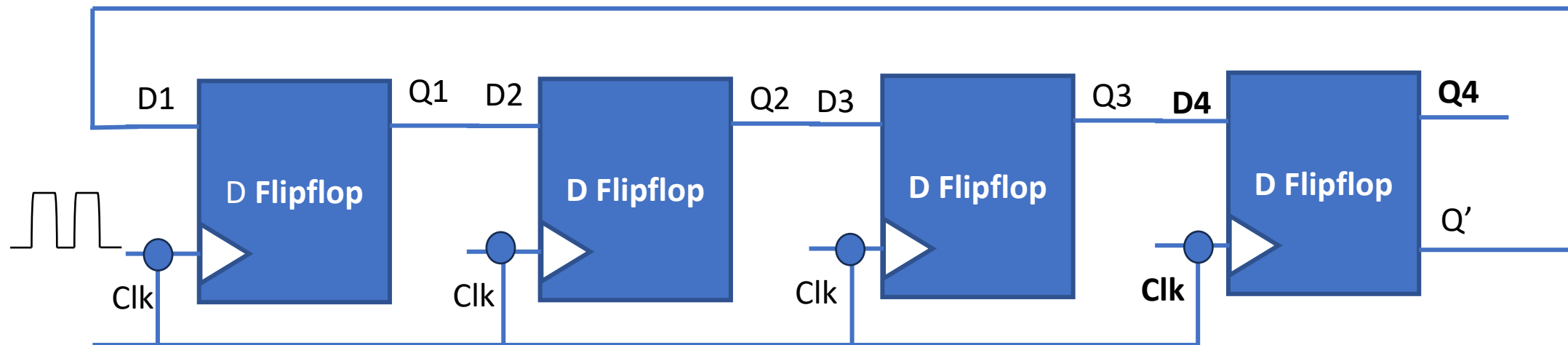
After the clock edge:

$Q1(t+1)$ becomes same as $Q3(t)$

$Q2(t+1)$ becomes inverse of $Q1(t)$

$Q3(t+1)$ becomes inverse of $Q2(t)$

Tail Ring Counter Four Stage



State
Transition
Table

Sequence No.	Q1	Q2	Q3	Q4
1	0	0	0	0
2	1	0	0	0
3	1	1	0	0
4	1	1	1	0
5	1	1	1	1
6	0	1	1	1
7	0	0	1	1
8	0	0	0	1
9	0	0	0	0